

9+ years building distributed real-time systems at scale — orchestrating 8,000+ concurrent devices across multi-tenant, multi-region deployments with zero-downtime failovers. Authored product specs, defined features from scratch, and led cross-functional teams from requirements through production delivery. Prolific open-source builder and early adopter of agent-driven development.

EXPERIENCE

Apra Labs — Associate Architect (2024–Present) · Senior SWE (2022–2024) · SWE (2019–2022) · AlienHu — Software Engineer (2017–2019)

SKILLS

< > LANGUAGES

C#
JavaScript / TypeScript
Python
Java
C / C++
PHP
Dart

FRONTEND

React / Redux
Angular
HTML5 Canvas
Electron

DATABASES

MySQL
MongoDB
Redis
CouchDB / PouchDB
Neo4j GraphDB

CLOUD & INFRA

Azure
AWS
GCP
Nginx

MOBILE

Flutter
Android
React Native
Ionic

REALTIME & AI

WebSockets / SignalR
WebRTC
MQTT / IoT
Machine Learning
NLTK

MORE

Node.js / Express
Agentic AI / MCP
Docker
CI / CD
Microservices
GraphQL

EDUCATION

B.E. in Computer Science & Engineering
SDMCET (2017), Karnataka

KEY PROJECTS

Video Surveillance System

2020 – Present

APRA LABS · ASSOCIATE ARCHITECT / LEAD

Architected an end-to-end cloud platform for orchestrating stable, low-latency live streams, configuration changes, and event captures for thousands of geo-distributed cameras across multiple tenants and regions. Engineered a scalable middleware layer with localized brokers and edge protocols for camera interfacing. Implemented reliable cross-region message routing via a distributed service bus and engineered real-time signaling and streaming state management across cloud boundaries for multi-region tenant isolation and seamless failovers. Led and mentored a cross-functional team — developers, QA, and support — driving stakeholder alignment across engineering, operations, and client teams from requirements through multi-region production rollouts.

Impact: Achieved zero-downtime regional failovers and sub-second stream switching, scaling to 8,000+ concurrent cameras while onboarding new enterprise tenants without additional infrastructure investment

Tech stack: C# (.NET), Azure Service Bus, SignalR, Azure Functions, Redis, WebRTC, MQTT, ONVIF, Milestone VMS, React, Redux, NSIS

Real-Time Multipurpose Dashboard

2023 – Present

APRA LABS · ASSOCIATE ARCHITECT / LEAD

As lead engineer, mentored a junior developer while designing and building a modular, content-heavy dashboard system from scratch — rendering hundreds of live-updating widgets and millions of concurrent data points on a complex, cross-building timeline without blocking the UI main thread. Engineered a custom HTML5 Canvas rendering pipeline bypassing traditional DOM limitations. Offloaded heavy calculations, stream parsing, and data manipulation to background Web Workers. Applied progressive data fetching, localized slice queries, and lazy evaluation of off-screen states to minimize memory overhead.

Impact: Sustained a locked 60 FPS across millions of live data points with zero browser memory leaks — powering 24/7 facility operations across multi-building security and access-control environments

Tech stack: C#, SignalR, SQL, Stored Procedures, React, Redux, HTML5 Canvas, Web Workers, TypeScript

E-Commerce Middleware & POS Platform

2017 – 2019

ALIENHU · SOFTWARE ENGINEER

Designed and developed a full-stack B2B point-of-sale application with a team, and single-handedly built the end-to-end B2C e-commerce mobile app and the shared sync middleware managing inventory, order notifications, and payment flows across fragmented POS networks. Engineered a fault-tolerant transaction queue with offline state management, eliminating double-selling and inventory conflicts with zero data loss during resync.

Tech: Node.js, CouchDB/PouchDB, Angular, Electron, React Native, Ionic, WebSocket

OPEN SOURCE & SHIPPED PRODUCTS

ToggleDev

Toggle Android Developer Options instantly — widget, Quick Settings tile, Banking App Shield, Remote Access Control, Automation, and more. 4.6★ rated · 6K+ users.
Kotlin, Android

Claude Shift

Align Claude Max 5-hour usage windows with your work schedule by scheduling strategic session pings during downtime. Cross-platform native OS schedulers.
TypeScript, Node.js

Terminal Studio

GPU-accelerated terminal multiplexer with built-in file browser, git diff viewer, code editor, and workspace persistence. Cross-platform, 120+ features, 15 themes.
Rust, wgpu, egui

Claude Lifecycle Core

Middleware hub for Claude Code plugins — register once, dispatch all lifecycle events to subscriber plugins. Eliminates per-plugin hooks.json boilerplate.
Node.js, JavaScript

AGENTIC AI TOOLS: GhostPM · Claude Comedy · AntiBluff · AntiClimax · Ingred · SplitGame · Claude Ticker
ALSO: CloseLink (Kotlin) · AfterCall Shield (Kotlin) · GitReg (Rust) · tTag (Go) · Workspace ID Pro (VS Code)